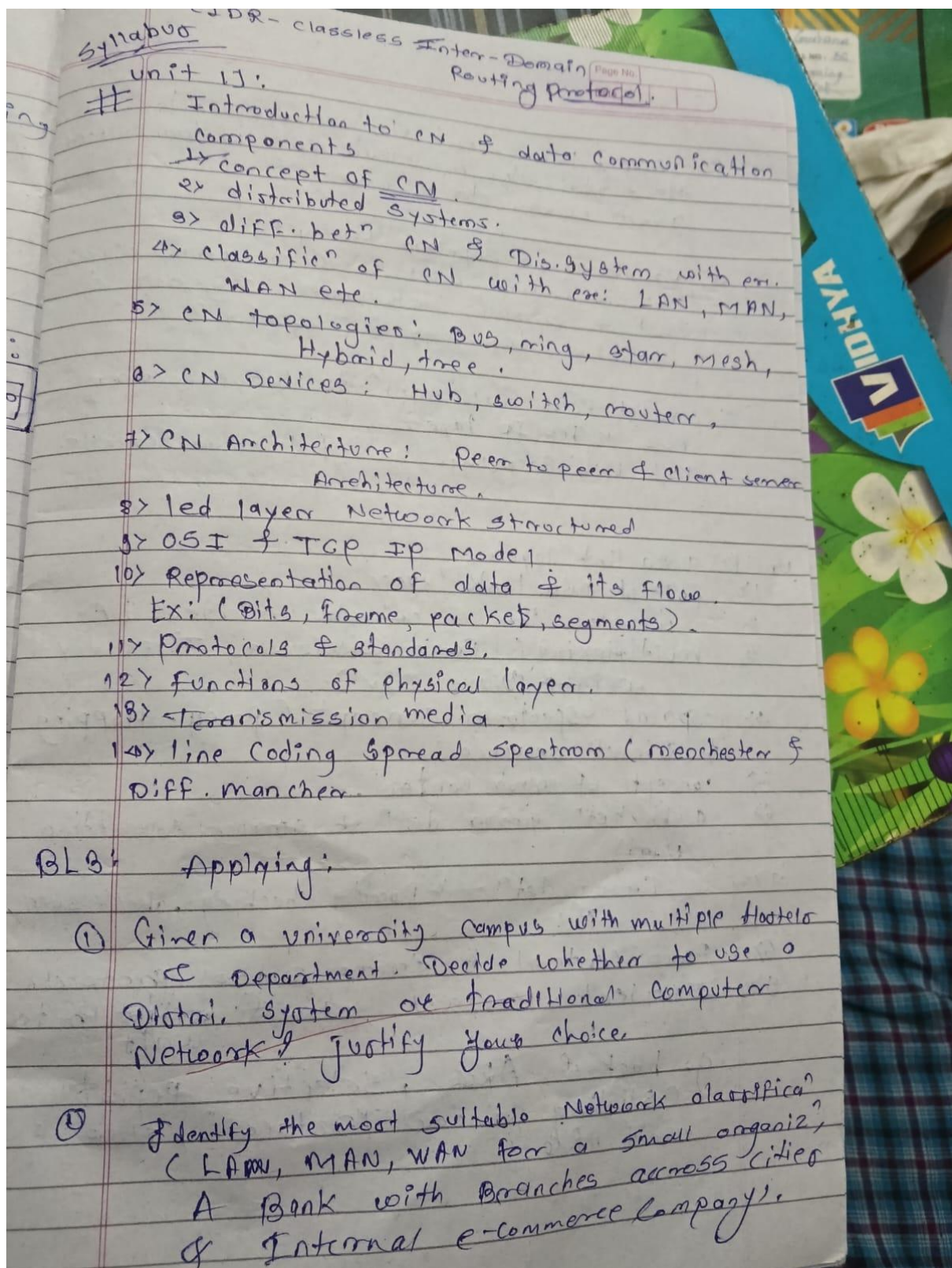




Q: Apply the concepts of layered architecture to explain how an email is transmitted from sender to receiver.

Answer:

The transmission of an email can be understood through the **OSI layered architecture**. Each layer performs a specific function and represents data in a particular form.

1. Application Layer (Data = Message / Email)

- **Role:** Sender composes an email using an application like Gmail/Outlook.
- **Protocols:**
 - **SMTP** (Simple Mail Transfer Protocol) → for sending mail.
 - **POP3/IMAP** → for retrieving mail at the receiver side.
- **Data Representation:** The email text, subject, and attachments are in **human-readable form**.

2. Presentation Layer (Data = Translated/Encoded Message)

- **Role:**
 - Converts data into a standard format (ASCII, EBCDIC, Unicode).
 - Handles **compression, encryption (SSL/TLS)**, and formatting of attachments using **MIME (Multipurpose Internet Mail Extensions)**.
- **Data Representation:** Standardized, encrypted, and possibly compressed **message data**.

3. Session Layer (Data = Same Message with Session Control)

- **Role:**
 - Establishes, maintains, and terminates communication between the email client and the mail server.
 - Provides dialog control and synchronization.
- **Data Representation:** Message with **session identifiers** to keep track of communication.

4. Transport Layer (Data = Segments)

- **Role:**
 - Breaks the email into **segments** for transmission.
 - Adds **TCP headers** (source & destination port numbers, sequence numbers, acknowledgments).
 - Ensures **reliable delivery, flow control, and error recovery**.
- **Data Representation:** **Segments** (each containing part of the email + TCP header).

5. Network Layer (Data = Packets / Datagrams)

- **Role:**

- Adds **IP addressing** (source IP = sender's machine, destination IP = receiver's mail server).
- Performs **routing** across networks so that data reaches the correct destination.
- **Data Representation: Packets** (TCP segment + IP header).

6. Data Link Layer (Data = Frames)

- **Role:**
 - Converts packets into **frames** for transmission over the physical medium.
 - Adds **MAC addresses** of sender's NIC and next-hop device (like a router).
 - Provides **error detection** using CRC.
- **Data Representation: Frames** (Packet + MAC header + trailer).

7. Physical Layer (Data = Bits)

- **Role:**
 - Converts frames into **binary 0s and 1s**.
 - Transmits signals (electrical, optical, or radio) over the physical medium (cables, fiber optics, Wi-Fi).
- **Data Representation: Raw bits** (signals representing 0s and 1s).

Receiver Side:

- The data flows **upward** through the layers.
- Physical → Data Link → Network → Transport → Session → Presentation → Application.
- At the **Application Layer**, the receiver finally reads the original **email message**.

✓ Conclusion:

In layered architecture, the email starts as a **message** at the Application Layer and moves down, being transformed into **segments → packets → frames → bits** for transmission. At the receiver end, the reverse happens, ensuring reliable delivery of the email.

Q: Identify the most suitable network classification for:

1. **A small organization**
 - **Answer: LAN (Local Area Network)**

- Reason: A small organization (like an office or startup) operates within a single building or campus → LAN provides **high speed, low cost, and resource sharing**.
-

2. A bank with branches across cities

- **Answer: WAN (Wide Area Network)**
 - Reason: Since branches are spread **across multiple cities**, a WAN is required.
 - WAN connects distant LANs using leased lines, satellite links, or the internet, ensuring **secure transactions and centralized databases**.
-

3. An internal e-commerce company

- **Answer: MAN (Metropolitan Area Network) or WAN (depending on scale)**
- If the company operates **within one city**, a **MAN** is most suitable (connects offices/warehouses across the city).
- If it operates **nationwide or globally**, a **WAN** is required to connect different locations and support online transactions.

Q: In a college/university, which system will you apply: Distributed System or Traditional Computer Network? Justify.

Answer:

In a university with multiple departments and hostels, the most suitable solution is a **Campus Area Network (CAN)**, which is essentially a **traditional computer network** interconnecting several **LANs**.

- Each building/department maintains its own LAN.
- The CAN connects these LANs through a high-speed backbone, allowing **resource sharing** (servers, printers, ERP, LMS).
- **Wi-Fi access** provides mobility for students and staff.

However, modern universities also rely on **distributed systems** for:

- Cloud services (online classes, storage, collaboration tools),
- Distributed databases (student records, research data),
- Scalability and fault tolerance.

Thus, a university network is a *hybrid system*:

- **Traditional CAN** provides physical connectivity and reliability.
- **Distributed systems** provide flexibility, scalability, and advanced services.

✅ **Conclusion:** Universities mainly use a **traditional CAN (traditional computer network)** as the base infrastructure, but enhance it with **distributed system elements** for modern requirements.

